What is RAG (Retrieval Augmented Generation)

How to use AI

RAG Enhances responses from LLMs

- Retrieval-augmented generation (RAG) is a technique for enhancing the accuracy and reliability of generative AI models with facts fetched from external sources.
- It fills the gaps in an LLMs knowledge, where it might not have a deeper understanding of a certain context or dataset

RAG Enables you to combine internal and external sources

- Take for example, you have a dataset of internal communications that is unique to your company.
- This information is not publicly available so the LLM would not have been trained on this data.
- By providing this in a RAG, you will let the LLM reference your data and cite it in order to provide a more accurate answer

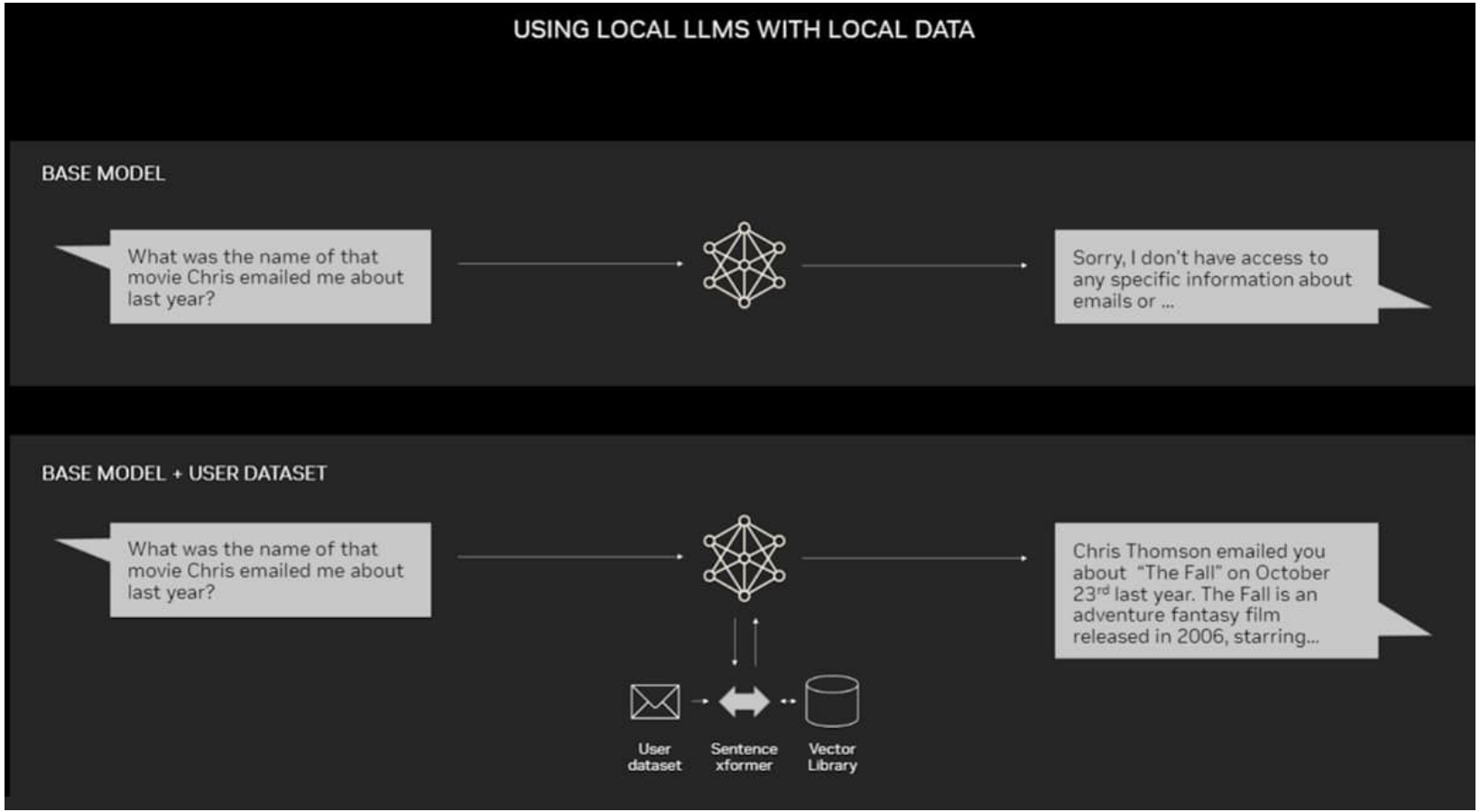
RAG enhances user trust

- RAG increases user trust, since regular LLMs don't include sources that arent publicly available. You can help guarantee a more relevant answer for the user that is querying the said LLM
- Data is not provided to a public LLM so user's data won't be surfaced in other people's queries.
- RAG can be implemented in 5 lines of code

```
from transformers import RagTokenizer, RagRetriever, RagTokenForGeneration

tokenizer = RagTokenizer.from_pretrained("facebook/rag-token-nq")
retriever = RagRetriever.from_pretrained("facebook/rag-token-nq", index_nam
model = RagTokenForGeneration.from_pretrained("facebook/rag-token-nq", retr

input_dict = tokenizer.prepare_seq2seq_batch("who holds the record in 100m

generated = model.generate(input_ids=input_dict["input_ids"])
```

Use cases of RAG

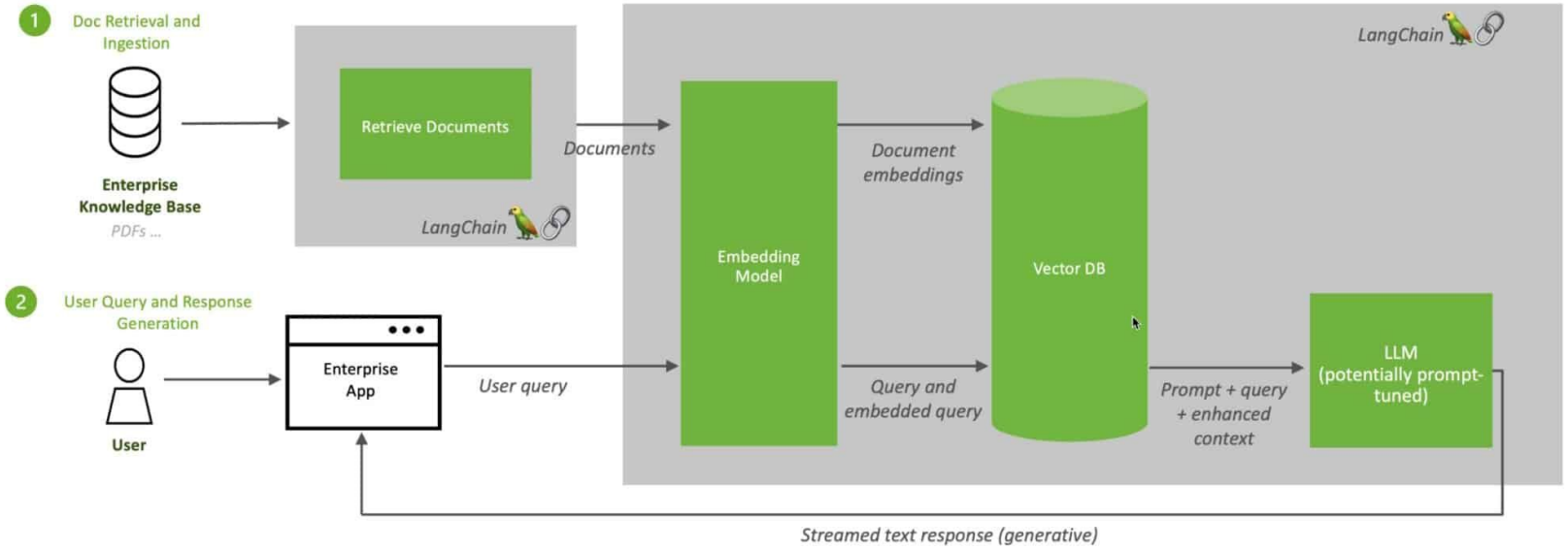
- Providing context to an LLM to patient medical records to answer medical questions
- Providing LLMs context to a users code base to fix bugs
- Answer customer support questions on a users internal documentation
- Acting as a virtual coworker that is able to resurface internal meeting notes and chatlogs
- And more


USING LOCAL LLMS WITH LOCAL DATA
BASE MODEL
What was the name of that movie Chris emailed me about last year?
Sorry, I don't have access to any specific information about emails or ...
BASE MODEL + USER DATASET
What was the name of that movie Chris emailed me about last year?
Chris Thomson emailed you about "The Fall" on October 23rd last year. The Fall is an adventure fantasy film released in 2006, starring...
User dataset
Sentence xformer
Vector Library

Process of performing RAG

- Keep sources in a data store
- Query for relevant sources when a user prompts the LLM
- Create and embedding to the LLM and input it into a vector database
- Query the LLM with embeddings
- Get response

Retrieval Augmented Generation (RAG) Sequence Diagram



Current Sources = Better Performance

- Your RAG application is only as good as your sources
- Keeping sources up to date is important to get relevant answers
- Consider building a system for keeping your sources up to date

Try RAG

- Notebook LM provides a user facing interface for using RAG
- https://notebooklm.google/